GOVERNMENT OF KHYBER PAKHTUNKHWA

KHYBER PAKHTUNKHWA TEXTBOOK BOARD PESHAWAR

OFFICIAL PRESCRIBED E-CURRICULUM TEXTBOOK

MATHEMATICS SCIENCE CLASS 9

CLASS / GRADE: 9th (Matric)

Subject: Mathematics | Language: English

AUTHENTIC CURRICULUM SYLLABUS & CHAPTER BLUEPRINTS

· Total Complete Curriculum Chapters: 12 Units

· Designed as per National Curriculum & KPK BISEP Standards

· Contains complete theory, conceptual exercises, MCQs & solutions

· Fully synchronized with ExamForge AI Examination Paper Generator

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UNIT 1: MATRICES AND DETERMINANTS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Matrices and Determinants.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 2: REAL AND COMPLEX NUMBERS

Class 9th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Real and Complex Numbers.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 3: LOGARITHMS AND LAWS OF LOGS

Class 9th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Logarithms and Laws of Logs.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 4: ALGEBRAIC EXPRESSIONS AND ALGEBRAIC FORMULAS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Algebraic Expressions and Algebraic Formulas.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 5: FACTORIZATION AND REMAINDER THEOREM

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Factorization and Remainder Theorem.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 6: ALGEBRAIC MANIPULATION (HCF & LCM)

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Algebraic Manipulation (HCF & LCM).

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 7: LINEAR EQUATIONS AND INEQUALITIES

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Linear Equations and Inequalities.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 8: LINEAR GRAPHS AND APPLICATIONS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Linear Graphs and Applications.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 9: INTRODUCTION TO COORDINATE GEOMETRY

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Introduction to Coordinate Geometry.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 10: CONGRUENT TRIANGLES & PARALLELOGRAMS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Congruent Triangles & Parallelograms.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 11: LINE BISECTORS AND ANGLE BISECTORS

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Line Bisectors and Angle Bisectors.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 12: PRACTICAL GEOMETRY - TRIANGLES

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Practical Geometry - Triangles.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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